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20 January 1988
SUPERSEDING
MIL-S-29410A(MC)
1 August 1987

MILITARY SPECIFICATION

SHELTER ASSEMBLY, 20 FOOT RIGID, GENERAL PURPOSE

This specification is approved for use by the United States Marine Corps, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense (DoD).

1. SCOPE

1.1 Scope. This specification establishes the requirements for the manufacture and acceptance of the Rigid Shelter Assembly. This shelter assembly is one of a family of rigid, walled shelters designed for Marine Corps expeditionary use.

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications, standards and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commandant of the Marine Corps, Code LMA-1, Washington D.C. 20380 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5411

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SPECIFICATIONS

MILITARY

MIL-C-5541	Chemical Conversion Coatings on Aluminum Alloys
MIL-H-21040	Honeycomb Materials, Water Migration Resistant Type, Structural Paper Base
MIL-S-29409	Shelter Assembly, Knockdown
MIL-S-29412	Shelter, Complexing Kit for

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-401	Sandwich Construction and Core Materials, General Test Methods
MIL-STD-810	Environmental Test Methods and Engineering Guidlines
MIL-STD-831	Test Reports, Preparation of
MIL-STD-907	Engineering and Design Criteria for Shelters, Expandable and Nonexpandable
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking and Waterproofing, with Appropriate Test Methods

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

PUBLICATIONS

MCO 4035.6	Surface Transportation Joint Procedures for Coding and Marking of DoD Owned Containers
MCO P4790.1	MIMMS Introduction Manual

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4.3 First article inspection

DRAWINGS

MARINE CORPS

DL80A5022A0001	Shelter, Rigid
DL80A5022A0357	Blackout Curtain Kit
DL80A5022A0164	Complexing Kit

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publicatons. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the non-government docuements which is current on the date of the solicitation.

INTERNATIONAL STANDARD

ISO 668	Dimensions and Ratings
ISO 1496/I	Series 1 Freight Container--Specification and Testing--Part I: General Cargo Containers

(Application for copies should be addressed to International Organization for Standardization, P.O. Box 56, 1211 Geneva 20, Switzerland.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM E-864-82	Surface Preparation of Aluminum Alloys to be Adhesively Bonded in Honeycomb Sandwich Panels
ASTM E-865-82	Structural Film Adhesives for Honeycomb Sandwich Panels
ASTM E-866-82	Corrosion-Inhibiting Adhesive Primer for Aluminum Alloys to be Adhesively Bonded in Honeycomb Sandwich Panels
ASTM E-874-82	Adhesive Bonding of Aluminum Facings to Nonmetallic Honeycomb Core for Shelter Panels

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

NATIONAL FIRE PROTECTION ASSOCIATION

NFPA 70-84 National Electric Code

(Application for copies should be addressed to the National Fire Protection Association 470 Atlantic Avenue, Boston, Massachusetts 02210.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article

3.1.1 Sandwich panel, large. One 4-ft (1.219-m) by 8-ft (2.438-m) aluminum face honeycomb sandwich panel fabricated in accordance with the requirements specified in 3.3.1 shall be subjected to the panel shear test in 4.8.1 and shall be subsequently cut into smaller specimens for other panel testing (see 4.8.2 and 4.8.8).

3.1.2 Assemblies. When specified in the contract or purchase order, sample shelter assembly(ies) shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Materials. Materials shall conform to the requirements specified herein and in Data Lists DL80A5022A0001, DL80A5022A0357, and DL80A5022A0164.

3.3 Design and construction

3.3.1 Sandwich panels. Sandwich panels shall be fabricated and inspected in accordance with the appropriate materials and process specifications listed in Data List DL80A5022A0001 and as follows:

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3.3.1.1 Sandwich panels and material. The following material requirements are the minimum necessary to insure the shelter assembly fully meets all requirements stated herein.

3.3.1.1.1 Sandwich panel shear strength. The adhesive bond between the panel facings and core shall be sufficient to allow the core to work to its failing shear stress. The composite sandwich panel shear strength at normal temperature (75° ± 5°F) shall be 80 psi (551,584 N/m^2) in the T_L direction parallel to the core ribbons. Testing shall be in accordance with 4.8.1.

3.3.1.1.2 Impact resistance. A 24-inch (in) (0.610-m) square sample of all exterior shelter panel(s) shall withstand a blow from a 70-lb (31.8-kilogram (kg)) steel cylinder (3 in (76.2 millimeter (mm)) in diameter with a hemispherical end), dropped from a height of 30 in (0.762 m) (16 in (0.406m) for 2.5# density honeycomb), impacting the hemispherical end without rupture to either skin and no delamination between skins and core outside a 3-in (76.2-mm) radius from the center of impact. Localized crushing of the core is allowable. Testing shall be in accordance with 4.8.2.

3.3.1.2 Surface preparation of aluminum alloys to be adhesively bonded in honeycomb shelter panels. Preparation of clean uniform surfaces of aluminum alloys suitable for formation of durable adhesive bonds to nonmetallic honeycomb materials in the manufacture of sandwich panels for the shelter assembly shall be in compliance with ASTM E-864-82. Honeycomb core material must comply with MIL-H-21040.

3.3.1.3 Structural film adhesives for honeycomb sandwich panels. Film adhesives used for bonding of honeycomb sandwich panels used in the shelter assembly shall be in accordance with ASTM E-865-82 and shall be suitable for forming bonds which can withstand long exposures to temperatures from -70°F (-56.7° Celsius (C)) to +160°F (+71.1°C) and also withstand the combinations of stress, temperature, and relative humidity expected to be encountered in service. The adhesives shall be used for bonding aluminum alloy facings to nonmetallic core, inserts, edge attachments and other components of a sandwich panel. The adhesives may be used for new production or depot repair. Honeycomb core materials will meet requirements of MIL-H-21040. The adhesive shall meet the requirements of the test specified in 4.8.22.

3.3.1.4 Corrosion-inhibiting adhesive primer for aluminum alloys to be adhesively bonded in honeycomb shelter panels. Primer shall be a sprayable, pigmented liquid primer for use on aluminum alloys which are to be adhesively bonded in the fabrication of honeycomb sandwich panels for the shelter assembly.

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When applied to a properly cleaned surface of aluminum alloy, the primer will impart corrosion resistance and form a surface suitable for structural bonding using adhesives complying with ASTM E-865-82 and for coating with shelter paint finishes. Corrosive-inhibiting adhesive primer shall comply with standards contained in ASTM E-866-82. All surfaces to be bonded will comply with this standard or meet performance standards using alternative chemical treatment processes to be specifically approved by the contracting agency.

3.3.1.5 Adhesive bonding of aluminum facings to nonmetallic honeycomb core for shelter panels. Materials, processes and quality controls to be used in the manufacture of adhesive-bond, aluminum-faced, nonmetallic, honeycomb-core sandwich panels for the shelter assembly shall comply with standards contained in ASTM E-874-82.

3.3.1.6 Core splice adhesive for honeycomb sandwich shelter panels. The adhesive shall be suitable for forming bonds which can withstand long combinations of stress, temperature, and relative humidity expected to be encountered in service. The adhesive shall be used for bonding core segments, bonding core segments to edge members, and integral members in a honeycomb sandwich panel. The adhesive shall not allow the transfer of water from one part of the panel to another either through or along the line of the adhesive.

3.3.2 Fastenings and joints. Welding, riveting and sealing of joints shall comply with the applicable standards and drawings in Data List DL80A5022A0001.

3.3.3 Panel flatness. Panel surfaces shall not be cupped or bowed in excess of 0.125 in (3.175 mm) when measured with a 48-in (1.219-m) long straightedge. Panels shall be tested in accordance with 4.8.3.

3.3.4 Interchangeability. Like demountable hardware and components of the shelter assembly shall be interchangeable.

3.3.5 Ease of operation. The demountable side panels and air conditioner panels shall permit repetitive installation and removal without binding or other loss of serviceability. Doors shall operate smoothly, maintaining adequate sealing at all times.

3.3.6 Forklift pockets. The shelter assembly shall have forklift pockets in accordance with the applicable drawings. The inner pair of pockets is intended to be used when the assembly is empty or lightly loaded while the outer pair of pockets is intended for use when the assembly is loaded and shall be capable of use with uniform loads up to and including 18,750 lbs (83,400 Newtons) gross weight including the weight of the assembly, without permanent deformation or abnormality. Testing shall be in accordance with 4.8.4.

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3.3.7 Blackout curtain kit. When specified (see 6.2), the blackout curtain kit shall be manufactured in accordance with Data Lists DL80A5022A0357. The blackout curtain shall prevent the exit of light from internal to the shelter assembly to external during entry or egress by personnel except for a small amount which will be visible at the bottom of the curtain. The blackout curtain shall be constructed from black nylon fabric, 2.2 ounces per square yard (74.6 grams (gm) per square meter) and coated with neoprene, 10.5 ounces per square yard (356 gm per square meter).

3.3.8 Electrical. The shelter assembly shall contain an electrical service entrance consisting of a 3-phase, 5-wire, wye-connected, 120/208-volt (V), 60-hertz (Hz), 60-ampere (A) circuit. The service shall be protected by a 3-phase, 60-A circuit breaker. Branch circuits to light fixtures and convenience outlets shall be protected by 15-A, single-phase circuit breakers. All wiring shall conform to NFPA 70-78 and shall be housed in an electrical metallic tubing conduit. The service entrance shall have one quick disconnect inlet (MS9055C32412P) and one quick disconnect outlet (MS9055C32412S) which are to be wired in parallel with phase rotation kept the same on input and output. All light fixtures, load center, distribution wiring and wiring devices shall be surface mounted. Eight surface mounted, 3-prong, 120-V, 15-A, 60-Hz duplex convenience outlets shall be provided. Eight 2-bar, 40-watt (W), rapid start, surface mounted fluorescent light fixtures shall be provided, switchable from a single switch mounted as shown on the applicable drawings. Testing shall be in accordance with 4.8.6.

3.3.9 Leveling. The shelter assembly shall have a device at each corner to indicate level when the floor is level to within 1/4 in in 6 ft (6.4 mm in 1.829 m). Leveling shall be accomplished utilizing jacks and corresponding fittings at the corners. The jacks and fittings shall withstand the force applied during the test specified in 4.8.7.

3.3.10 Fire resistance. The shelter assembly shall be fire resistant and shall meet the test requirements of 4.8.8.

3.3.11 Coast Guard certification. The shelter assembly shall be certified as a 8-ft X 8-ft X 20-ft, 1C container for its maximum rated load in accordance with the applicable Coast Guard regulation for such containers for all types of transport, including motor carrier (truck), sea, railroad, and air. Qualification shall be in accordance with the requirements of ISO 1496/I. At a minimum, testing shall be in accordance with 4.8.9 through 4.8.17.

3.3.12 Complexing. The shelter assembly shall be capable of complexing with any other like shelter or the knockdown shelter assembly, built in accordance with MIL-S-29409 side-to-side to

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form a structure 20 ft (6.1 m) deep by any multiple of 8 ft (2.44 m) wide. To adequately seal the complexed shelters one (or more) complexing kits built in accordance with MIL-S-29412 will be required. Testing shall be in accordance with 4.8.18.

3.3.13 Metal fabrication. Metal used in the fabrication of equipment shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Shearing and punching shall be done neatly and accurately. Corners shall be square and true. All bends of a major character shall be made with controlled means in order to insure uniformity of size and shape.

3.3.14 Facing material. Panel facings shall be one piece seamless aluminum alloy skins of a material and thickness specified on the applicable drawings. All edges of aluminum faced honeycomb panels shall be completely sealed to prevent water intrusion between the laminates.

3.3.15 Bolted and riveted connectors. Bolt and rivet holes shall be accurately punched or drilled. All bolt and rivet connectors shall have the burrs removed. Washers, lockwashers, or locknuts shall be provided where necessary and all bolts, nuts and screws shall be tight. Rivet heads, when not countersunk or flattened, shall be of uniform size and shape for the same diameter rivet, concentric with the rivet holes, and in full contact with the surface of the members.

3.3.16 Welding. The surface of the parts to be welded shall be free from rust, scale, paint, grease or other foreign matter. Weld penetration shall be such as to provide transference of maximum design stress through the base metal juncture. Fillets shall be as specified on the applicable drawings.

3.3.17 Castings. Castings shall be sound and free from patching, misplaced coring, warping or defects which might render the castings unsound for use.

3.3.18 Seals. Seals shall be installed so that the fit within the retaining track is snug and continuous contact is maintained with mating seal faces. Mitered end junctions of vertical and horizontal seal runs shall be accurate, and gaps shall be sealed as specified on the applicable drawings. The resulting shelter assembly shall have an air volume leak rate of less than 200 standard cubic ft/minute (scfm) (60.96 standard m^3/min) with an internal pressure of 1.2 in H_2O (30.48 mm H_2O).

3.4 Maintainability. The shelter assembly has been designed so that as much of the maintenance, as possible, may be performed as far forward (preferably by the using unit), as feasible, within

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the Marine Corps maintenance structure, in accordance with MCO P4790.1.

3.5 Transportability

3.5.1 Stacking. The shelter assembly shall withstand the requirements of the Coast Guard intermodal transportation requirements for land, sea and air transport as a 20-ft, 1C shipping container with a gross weight of 15,000 lbs (66,720 Newtons). Testing shall be in accordance with 4.8.19.7.

3.5.2 Railroad transportability. The shelter assembly shall withstand without damage the shocks normally induced by transport by the railroads. Testing shall be in accordance with 4.8.20.3.

3.6 Environmental requirements

3.6.1 Rainfall. The shelter assembly shall withstand rainfall up to and including 5 in per hour (in/hr) (127 millimeters per hour (mm/hr)) without leakage at any location. Testing shall be in accordance with 4.8.19.1.

3.6.2 Temperature-humidity. In storage, the shelter assembly shall be capable of withstanding exposure to temperatures of -70°F (-56.7°C) to 160°F (71.1°C). In transit, the shelter assembly shall be capable of withstanding exposure to temperatures of -65°F (-53.9°C) to 160°F (71.1°C) with personnel access at low end of range. Operational temperature of shelter assembly shall be -65°F (-53.9°C) to 125°F (51.7°C) plus solar load (see 3.6.8). The shelter assembly shall withstand daily exposure of up to 97 percent relative humidity for 20 hours and exposure of 100 percent relative humidity (with condensation) for 4 hours. Testing shall be in accordance with 4.8.19.2 and 4.8.19.3.

3.6.2.1 Thermal shock. The shelter assembly shall withstand sudden temperature changes from 160°F (71.1°C) to -70°F (-56.7°C) to 160°F (71.1°C) without separation, delamination, cracks or degradation of physical properties. Testing shall be in accordance with 4.8.19.4.

3.6.2.2 Heat transfer. The shelter assembly shall maintain an overall average heat transfer coefficient less than or equal to 0.35 British thermal unit (BTU)/hr-Ft^2-°F (1.99 W/m^2-°K). Testing shall be in accordance with 4.8.19.5.

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3.6.3 Snow and ice. The shelter assembly shall be capable of functioning in accordance with the requirements of this specification during periods of snow and ice storms. The shelter shall inhibit the collection of snow and ice so that its primary function is not degraded. The shelter shall be capable of snow loads of up to 40 lbs per square ft (1,915 N/m^2) in addition to its internal load.

3.6.4 Wind. The shelter assembly shall be capable of withstanding steady state wind speeds of up to 65 mph (104 km/hr) and gusts up to 120 mph (192 km/hr), static load on the door of 15 lbs per square ft (718 N/m^2) and a dynamic load on the door of 50 lbs (22.7 kg) applied to the midpoint of the locking edge acting in a horizontal plane perpendicular to the front panel of the door in both directions, the force increased by dropping the weight (50 lbs) a distance of 12 in (0.305 m). Testing shall be in accordance with 4.8.19.6. In winds in excess of 55 mph (88 km/hr), ground anchors are required.

3.6.5 Roof load. The shelter assembly shall withstand a uniform load of 40 lbs per square foot (1,915 N/m^2) over the entire roof of the shelter without permanent damage and a concentrated load of 660 lbs (2,936 Newtons) over an area 24 in x 12 in (0.610 m by 0.305 m) without permanent damage rendering the shelter assembly unusable. Testing shall be in accordance with 4.8.19.8 and 4.8.19.9.

3.6.6 Sand and dust. The shelter assembly shall withstand the sand and dust effects on all external moving parts with particle concentrations of 0.000132 pounds per cubic foot ($lb/cu\text{-}ft^3$) (2.19 gm/m^3), with wind velocity of 1,750 ± 250 ft/min (533 ± 76.2 m/min) without degradation. Such particles shall range in size from 0.00294 in (0.150 mm) up to 0.0197 in (1.0 mm) with relative humidity less than 23 percent. Testing shall be in accordance with 4.8.19.10.

3.6.7 Sunshine. The shelter assembly shall withstand the effects of ultraviolet radiation (sunshine) without significant degradation or effect on serviceability of the assembly's components or materials during the service life of the shelter assembly. Testing shall be in accordance with 4.8.19.11.

3.6.8 Solar load. The shelter assembly shall withstand a solar heat load on its roof sufficient to raise the outer skin to at least 200°F (93.3°C) without any evidence of delamination or permanent deformation. Testing shall be in accordance with 4.8.19.12.

3.6.9 Marine environment. The shelter assembly shall be fully serviceable when exposed to sea-salt fallout equivalent to 12.5

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pounds/acre/year (1.4 gm/m^2/year). All fasteners, jacks, seals, or other hardware shall show no evidence of corrosion or degradation when tested as required by 4.8.19.13.

3.6.10 Drop shock. The shelter assembly shall withstand without damage the following drop shocks when loaded to the maximum payload weight uniformly spread over the floor. After each test, there shall be no permanent deformation, buckling, delamination sealer separation or structural failures to any part of the shelter assembly. Doors and covers shall open and close to their full extent without binding.

3.6.10.1 Flat drop. The shelter assembly shall withstand a flat drop of 12 in (300 mm) from the ground onto a concrete surface. Testing shall be in accordance with 4.8.20.1.

3.6.10.2 Rotational drop. The shelter assembly shall withstand rotational drops, with one edge placed on a 4-in (102-mm) block and the opposite edge raised 12 in (305 mm) and allowed to free fall onto a concrete surface with the 4-in (102-mm) block acting as a pivot. The shelter assembly shall withstand this drop on each bottom edge. Testing shall be in accordance with 4.8.20.2.

3.6.11 Blackout. The shelter assembly shall be light-tight with the door closed. The shelter assembly shall have built in inserts for blackout curtains and associated electrical switches. When equipped with a blackout curtain kit built in accordance with DL80A5022A0357, light shall not escape during entry/egress except for a small amount of light which will be visible at the bottom of the curtain where it meets the floor. Testing shall be in accordance with 4.8.21.1.

3.6.12 Floor loading. The floor of the shelter assembly shall be capable of withstanding uniform floor loads of 65 lb/ft^2 (3112 N/m^2) over the entire area of the floor and a uniform concentrated load of 2,000 lbs (8,896 Newtons) over a 4-ft^2 (0.372-m^2) area. Testing shall be in accordance with 4.8.21.2.

3.7 Dimensional characteristics. The exterior envelope of the shelter assembly shall comply with ISO Shipping Container Standards as defined in ISO 1496/I for a 20-ft (6.096-m) container. The applicable drawings fully describe all pertinent dimensions and tolerances. Should any dimensions and tolerances result in a component or assembly which will not meet the requirements of this specification, the supplier is responsible for fabrication techniques which will prevent the occurrence of such dimensions and tolerances. Testing shall be in accordance with 4.8.5.

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3.8. Weight. The tare weight of the shelter assembly including all permanent accessories shall be 3,750 lbs ± 5 percent (16,680 Newtons ± 5 percent). The gross weight shall not exceed 15,000 lbs (66,720 Newtons). The actual shelter weight shall be stamped on the nameplate.

3.9 Finish

3.9.1 Surface preparations. All surfaces shall be prepared for painting as specified on the applicable drawings. All aluminum alloy materials shall be protected by a chemical film conforming to MIL-C-5541 even if a final finish is not applied.

3.9.2 Priming. All surfaces to be painted shall be primed as specified on the applicable drawings. Panel edge members and frames shall be processed in accordance with ASTM E-864-82 and protected from corrosion by the chemical conversion coating in accordance with MIL-C-5541, class 3, or a corrosion inhibiting adhesive primer in accordance with ASTM E-866-82.

3.9.3 Final finish. Painted surfaces shall be as specified on the applicable drawings.

3.10 Marking. Marking shall be in accordance with Drawing No. 80A5022A0336 and in compliance with coding, marking and Convention for Safe Containers (CSC) plating requirements established in MCO 4035.6.

3.10.1 Identification. Each shelter assembly shall have the following information provided on a serial plate permanently affixed to the shelter assembly in accordance with MIL-STD-130:

Manufacturer:
Model:
Type:
Serial No.: (As assigned by the Procuring Agency)
Weight:

3.10.2 Certification. Each shelter assembly shall carry the appropriate certification (CSC safety approval plate) from the approval authority as to its capability to act as a shipping container.

3.11 Workmanship. All parts, components and assemblies of the shelter assembly, including casting, forgings, molded parts, stampings, seals and sealing agents, machined surfaces and welded parts shall be clean and free from any defects that will reduce the capability of the shelter assembly to meet the requirements specified herein. External surfaces shall be free from burs, slag,

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sharp edges and corners, except where sharp edges and corners are required. Any components and assemblies which have been repaired or modified to overcome deficiencies or deficient components or assemblies which are determined to be used "as is" shall not be used unless authorized by the contracting agency.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspections specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set fourth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)

b. Quality conformance inspection (see 4.5)

c. Individual inspection (see 4.6)

d. Daily inspection (see 4.7)

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.4, 4.5, 4.6, and 4.7.

4.4 First article inspection. Prior to beginning production of the shelter assemblies, the contractor shall fabricate one 4-ft x 8-ft honeycomb shelter panel and two first article

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shelter assemblies and subject them to the tests specified in Table I.

4.5 Quality conformance inspection. From each shipping lot as defined in the contract, a single shelter assembly shall be chosen at random by the Government and subjected to the tests specified in Table I. Failure of any test shall be cause for inspection of the entire lot at the contractor's expense for similar defects. Defective shelter assemblies shall not be delivered without correction of the deficiency and then only with the permission of the contracting officer.

4.6 Individual inspection. Each shelter assembly shall be subjected to the inspections and tests specified in Table I. Failure to meet the requirements of the drawings and this specification shall be cause for rejection. No shelter assembly shall be reworked and retested without the expressed permission of the contracting agency.

4.7 Daily inspections. During the manufacture of the honeycomb panels, daily inspections shall be performed (see 4.8.22).

4.8 Inspections and tests.

4.8.1 Sandwich panel shear. Five preproduction sandwich panel shear specimens shall be subjected to MIL-STD-401, Test Method 5.2.4.1. The specimens shall be deemed acceptable if a shear stress in excess of 80 psi (551,584 N/m^2) in the T_L direction parallel to the core ribbons is achieved. The contracting agency reserves the right to require testing of production panels to the same criteria.

4.8.2 Impact resistance. A 24-in by 24-in (0.610-m by 0.610-m) sample of each exterior shelter panel, roof and floor shall be tested in accordance with MIL-STD-907, Par 5.3.14.1 except for 2.5# density honeycomb which shall be tested using a height of 16 in. (0.406M).

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TABLE I. Tests

INSPECTION	REQ PARA	TEST PARA	1ST ART	QUAL CONF	IND
Sandwich panel shear	3.3.1.1.1	4.8.1	X		
Impact resistance	3.3.1.1.2	4.8.2	X		
Panel flatness	3.3.3	4.8.3	X	X	
Lift by fork pockets	3.3.6	4.8.4	X		
Dimensional	3.7	4.8.5		X	X
Electrical	3.3.8	4.8.6	X	X	X

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TABLE 1. TESTS - CONTINUED

INSPECTION	REQ PARA	TEST PARA	1ST ART	QUAL CONF	IND
Leveling	3.3.9	4.8.7	X	X	X
Fire resistance	3.3.10	4.8.8	X		
Transverse and longitudinal rigidity	3.3.11	4.8.9	X		
Lifting from top	3.3.11	4.8.10	X		
Lifting from bottom	3.3.11	4.8.11	X		
Aerodynamic end	3.3.11	4.8.12	X		
Aerodynamic side	3.3.11	4.8.13	X		
Endwall strength	3.3.11	4.8.14	X		
Sidewall strength	3.3.11	4.8.15	X		
Longitudinal restraint	3.3.11	4.8.16	X		
Coast Guard certification	3.3.11	4.8.17	X	X	X
Complexing	3.3.12	4.8.18	X		
Watertightness	3.6.1	4.8.19.1	X		
Humidity resistance	3.6.2	4.8.19.2	X		
Temperature range	3.6.2	4.8.19.3	X		
Temperature shock	3.6.2.1	4.8.19.4	X		
Heat transfer	3.6.2.2	4.8.19.5	X		
Door load	3.6.4	4.8.19.6	X		
Stacking	3.5.1	4.8.19.7	X		
Roof load, uniform	3.6.5	4.8.19.8	X		
Roof load, concentrated	3.6.5	4.8.19.9	X		
Blowing sand	3.6.6	4.8.19.10	X		
Sunshine	3.6.7	4.8.19.11	X		
Solar load	3.6.8	4.8.19.12	X		
Marine atmosphere	3.6.9	4.8.19.13	X		
Drop shock	3.6.10.1	4.8.20.1	X		
Rotational drop	3.6.10.2	4.8.20.2	X		
Railroad transport	3.5.2	4.8.20.3	X		
Blackout	3.6.11	4.8.21.1	X		
Floor load	3.6.12	4.8.21.2	X		
Airtightness	3.3.18	4.8.21.3	X	X	
Adhesive Shear	3.3.1.3	4.8.22			

4.8.3 Panel flatness. The shelter assembly shall be tested in accordance with MIL-STD-907, PAR 5.3.4.1.

4.8.4 Lift by fork pockets. The shelter assembly under test shall be uniformly loaded to a gross weight of 18,750 lbs (83,400 Newtons). The shelter assemblies shall be supported by two horizontal bars, each 8 in (20 mm) wide, all the way through the outer fork pockets of the bottom assembly with sufficient length protruding outside the walls to attach lifting cables. The shelter assemblies shall then be lifted clear of the ground, supported for

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5 minutes and then lowered to the ground. Upon completion of the test, the shelter assemblies shall show neither permanent deformation nor abnormality which will render them unsuitable for use.

4.8.5 Dimensional. The length (L), width (W), height (H), and diagonals (K1 and K2) shall be measured and recorded for each shelter assembly. The requirements of Tables 3 and 4 of ISO 668 shall be satisfied for a 20-ft, 1C container.

4.8.6 Electrical. The shelter assembly under test shall be connected to a 120/208-V, 3-phase, 5-wire, 60-A electrical power source. The lights shall be turned on and operate properly. Each convenience outlet shall be checked with a voltmeter to assure 120 V between line and neutral and line and ground, and zero voltage between neutral and ground. Each lamp shall be visually checked for its protective shield. During the first article inspection, only the phase rotation of the output connector shall be verified as the same as the input phase rotation using a 3-phase motor and assuring that its direction of rotation is identical.

4.8.7 Leveling. The shelter assembly shall be leveled such that all level indicators are centered. The floor shall be checked using a 6-ft (1.830-m) long level for both lateral and longitudinal level at each end of the shelter assembly. Any indication of slant in excess of 1/4 in (6.4 mm) over 6 ft (1.830 m) shall be cause for rejection. The shelter assembly shall be subjected to tests in MIL-STD-907, PAR 5.3.9.1, except the height is limited to 18 in (0.457 m), during first article and quality conformance inspection. Any level which cannot be centered shall be cause for rejection.

4.8.8 Fire resistance. Two 12-in by 12-in (300-mm by 300-mm) specimens of a production sandwich panel shall be subjected to tests in MIL-STD-907, PAR 5.3.1.1.

4.8.9 Transverse and longitudinal rigidity. The shelter assembly under test shall be placed on four level supports, one under each corner fitting and shall be restrained against lateral and vertical movement by means of anchor devices, acting through the bottom corner fittings. Lateral restraint shall be provided only at a bottom corner fitting diagonally opposite to and in the same end frame as a top corner fitting to which force is applied. When testing the two end frames separately, vertical restraint shall be applied only at the end frame under test. A force of 33,600 lbs (149,460 N) shall be applied either separately or simultaneously, to each of the top corner fittings on one side of the shelter assemblies under test in lines parallel to both the base and to the planes of the ends. The forces shall be applied first towards, then away from, the top corner fittings. Upon completion of the test, the shelter

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assemblies shall show neither permanent deformation nor abnormality which will render them unsuitable for use, and shall be subjected to the test in 4.8.5. This test shall be repeated with the forces acting parallel to the long side of the shelter assembly.

4.8.10 Lifting from top. The shelter assemblies under test shall have a load uniformly distributed over the floor in such a way that the combined weight of the shelter assembly and the test load is equal to 37,500 lbs (166,800 Newtons). It then shall be carefully lifted from all four top corners in such a way that no significant acceleration or deceleration forces are applied. The shelter assemblies shall be suspended by the lifting mechanism for 5 minutes and then carefully lowered to the ground. Upon completion of the test, the shelter assemblies shall show neither permanent deformation nor abnormality which will render them unsuitable for use, and shall be subjected to the test in 4.8.5.

4.8.11 Lifting from bottom. The shelter assembly under test shall have a load uniformly distributed over the floor in such a way that the combined weight of the shelter assembly and the test load is equal to 37,500 lbs (166,800 Newtons). It shall then be carefully lifted from the side apertures of all four bottom corner fittings in such a way that no significant acceleration or deceleration forces are applied. Lifting forces shall be applied at 45° from the vertical, parallel to the long sides of the shelter assemblies. In each case, the line of action of the lifting force and the outer face of the corner fitting shall be no further apart than 1.5 in (38 mm). The lifting shall be carried out in such a manner that the lifting devices bear on the bottom corner fittings only. The shelter assemblies shall be suspended for 5 minutes and then lowered carefully to the ground. Upon completion of the test, the shelter assemblies shall show neither permanent deformation nor abnormality which will render them unsuitable for use, and shall be subjected to the test in 4.8.5.

4.8.12 Aerodynamic end. Place the shelter assembly on one end supported by the four corner fittings. Uniformly load the upper endwall with 5,120 lbs (22,773 Newtons) (80 pound-force per square foot (psf)). Allow the upper endwall to remain loaded for 5 minutes. Repeat the test for the opposite endwall. The shelter assembly when righted to its normal sitting position after completion of this test there shall be no evidence of damage, and shall be subjected to the test in 4.8.5.

4.8.13 Aerodynamic side. Place the shelter assembly on its side supported by the four corner fittings. Uniformly load the upper sidewall with 12,000 lbs (53,376 Newtons) (80 psf). Allow the upper sidewall to remain loaded for 5 minutes. When the shelter

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assembly is righted to its normal sitting position after completion of this test, there shall be no evidence of damage, and shall be subjected to the test in 4.8.5.

4.8.14 Endwall strength. Each end of the shelter assembly under test shall be loaded to 9,225 lbs (41,033 Newtons), uniformly distributed over the entire endwall surface. The endwall shall be supported to allow free deflection of the wall. Upon completion of the test, the shelter assembly shall show neither permanent deformation nor abnormality which will render it unsuitable for use, and shall be subjected to the test in 4.8.5.

4.8.15 Sidewall strength. One sidewall of each shelter assembly under test shall be loaded to 9,225 lbs (41,033 Newtons), uniformly distributed over the entire sidewall surface. The sidewall shall be supported to allow free deflection of the wall. Upon completion of the test, the shelter assembly shall show neither permanent deformation nor abnormality which will render it unsuitable for use, and shall be subjected to the test in 4.8.5.

4.8.16 Longitudinal restraint. The shelter assembly under test shall have a load uniformly distributed over the floor in such a way that the gross weight of the shelter assembly is 15,000 lbs (6,804 kg). The shelter assembly shall then be restrained by rigid anchor points, two on one end only. These anchors shall be through the bottom apertures of the bottom corner fittings. A force of 30,000 lbs (133,400 Newtons) (twice the gross weight of the shelter) shall be applied horizontally to the shelter assembly through the bottom apertures of the other two corner fittings, first towards, and then away from the anchor points. Upon completion of the test, the shelter assemblies shall show neither permanent deformation nor abnormality which will render them unsuitable for use, and shall be subjected to the test in 4.8.5.

4.8.17 Coast Guard certification. The shelter assembly shall be tested for Coast Guard certification. If there is any damage to the shelter assembly due to these tests, it will be the responsibility of the contractor to repair the damage or make any correction necessary for the shelter assembly to pass Coast Guard certification at his own expense. A certificate of Coast Guard approval shall be furnished and Coast Guard certification plates shall be attached to each delivered shelter assembly.

4.8.18 Complexing. The two first article shelter assemblies shall be complexed side-to-side using the complexing kit conforming to MIL-S-29412. Any failure to meet, match and otherwise fit together shall be cause for rejection.

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4.8.19 Environmental tests

4.8.19.1 Watertightness. The shelter assembly shall be subjected to watertightness test (panels) in accordance with MIL-STD-907, PAR 5.3.16.1. The shelter assembly, without the use of any external sealing, caulking, taping, etc., shall show no evidence of leakage.

4.8.19.2 Humidity resistance. The shelter assembly shall be subjected to MIL-STD-810, Method 507.1, Procedure II, excluding steps 2-4, 7, and 8. After cycling, the shelter assembly shall be examined and shall show no evidence of delamination, cracking, corrosion or deterioration of any part of the shelter assembly. The shelter assembly shall be energized electrically and the lights shall function normally.

4.8.19.3 Temperature range. The shelter assembly shall be cold soaked in a mechanically refrigerated cold chamber at -70°F (-56.7°C) for a minimum of 16 hours. The chamber temperature shall then be raised to -65°F (-53.9°C). The shelter assembly doors shall be opened and closed (both the standard door and the emergency escape hatch). Any malfunction shall be recorded and reported. The shelter assembly shall then be visually examined both internally and externally for material degradation. There shall be no damage to seals or other components and all hardware shall operate as specified during and after completion of this test.

4.8.19.4 Temperature shock. A representative shelter panel specimen measuring 4 ft (1.22 m) by 8 ft (2.44 m) shall be subjected to MIL-STD-810, Method 503.1. There shall be no separation, cracks, delamination or degradation of physical properties after completion of this test.

4.8.19.5 Heat transfer. The shelter assembly shall be subjected to heat transfer test in accordance with MIL-STD-907, PAR 5.2.8.1. The overall heat transfer coefficient shall be determined and the requirements of 3.6.2.2 shall be satisfied.

4.8.19.6 Door loads. The standard personnel door on the shelter assembly under test shall be subjected to tests specified in MIL-STD-907 PAR 5.3.7.1 and PAR 5.3.7.2 for static and wind gust loading. Any evidence of unbonded components, damage to hardware, or improper open door operation shall constitute failure of this test.

4.8.19.7 Stacking. The shelter assemblies under test shall be placed on four level pads, one under each bottom corner fitting. The pads shall be centered under the fittings and be substantially of the same plan dimensions as the fittings. The shelter assemblies

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under test shall then be subjected to a vertical force equivalent to a load of 100,800 lbs (45,773 kg), applied for 5 minutes simultaneously on each pair of end fittings in such a manner that the planes of application of the forces and the support of the shelter assemblies remain horizontal and unchanged during the test. The forces shall be applied through a corner fitting or a pad of the same plan area as a corner fitting. Each pad shall be offset in the same direction by 1 in (25.4 mm) laterally and 1.5 in (38 mm) longitudinally. Upon completion of the test, the shelter assemblies shall show neither permanent deformation nor abnormality which will render them unsuitable for use, and shall be subjected to the test in 4.8.5.

4.8.19.8 Roof load, uniform. The shelter assembly under test shall be placed on the ground and leveled. The roof of the shelter assembly shall then be loaded uniformly with a weight of 6,400 lbs (2,903 kg) (40 lbs/square foot) and allowed to remain loaded for a period of 5 minutes. Upon completion of the test, the shelter assembly shall show neither permanent deformation nor abnormality which will render it unsuitable for use, and shall be subjected to the test in 4.8.5.

4.8.19.9 Roof load, concentrated. A load of 660 lbs (300 kg) shall be uniformly distributed over an area of 12 in by 24 in (300 mm by 600 mm) located at the weakest area of the rigid roof of the shelter assembly (centered on the roof). This load shall be allowed to remain there for a period of 5 minutes. Upon completion of the test, the shelter assembly shall show neither permanent deformation nor abnormality which will render it unsuitable for use, and shall be subjected to the test in 4.8.5.

4.8.19.10 Blowing sand. The shelter assembly shall be subjected to blowing sand test in accordance with MIL-STD-907, PAR 5.2.9.1. There shall be no damage to seals or other components and all hardware shall operate as specified herein upon completion of this test.

4.8.19.11 Sunshine. Representative samples of all gaskets and seals, as well as a 12-in by 12-in (300-mm by 300-mm) panel sample, shall be subjected to MIL-STD-810, Method 505.1, Procedure I. Upon completion of this test, the samples shall be examined and any evidence of degradation of physical properties shall constitute failure of this test.

4.8.19.12 Solar load. The shelter assembly shall be subjected to solar load test-assembled shelter in accordance with MIL-STD-907, PAR 5.2.11.1. Upon completion of the solar load test, the roof panels shall be examined and any evidence of delamination or deformation will constitute failure.

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4.8.19.13 Marine atmosphere. One representative sample of each fastener, jack, seal, panel, and other hardware which will be exposed to the atmosphere in the operation and storage modes shall be finished in accordance with the applicable drawings and subjected to tests as described in MIL-STD-810, Method 509.2, except that the salt solution shall have a concentrate of 5 percent and the exposure period shall be 96 hours. The test items shall show no evidence of corrosion upon completion of the test.

4.8.20 Shock tests

4.8.20.1 Drop shock. The shelter assembly shall be uniformly loaded to the rated gross weight of 15,000 lbs (66,720 Newtons) and lifted 12 in (300 mm) above a flat concrete surface. The distance between the concrete surface and each lower corner fitting of the shelter assemblies under test shall not vary more than 0.5 in (12.7 mm) at the time of the drop. The shelter assemblies shall then be allowed to free fall to the concrete surface. Upon completion of the test, the shelter assemblies shall show neither permanent deformation nor abnormality which will render them unsuitable for use, and shall be subjected to the test in 4.8.5. This test shall be repeated with a single shelter in its shipping configuration.

4.8.20.2 Rotational drop. The shelter assembly shall be uniformly loaded to the rated gross weight of 15,000 lbs (66,720 Newtons) and with one edge supported 4 in (102 mm) above a flat concrete surface. The opposite edge shall be lifted a distance of 12 in (300 mm) above the concrete surface and allowed to free fall onto the surface. This procedure shall be repeated until each edge has been allowed to free fall. Upon completion of the test, the shelter assemblies shall show neither permanent deformation nor abnormality which will render them unsuitable for use, and shall be subjected to the test in 4.8.5. This test shall be repeated with a single shelter in its shipping configuration.

4.8.20.3 Railroad transport. The shelter assembly shall be uniformly loaded to the rated gross weight of 15,000 lbs (66,720 Newtons) and subjected to MIL-STD-810, Method 516.2, Procedure VI, with impact speeds between 9 mph (14.5 km/hr) and 10 mph (16.1 km/hr) in both directions. The shelter assemblies shall be subjected to two impacts in each direction. Upon completion of the test, the shelter assemblies shall show neither

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permanent deformation nor abnormality which will render them unsuitable for use, and shall be subjected to the test in 4.8.5.

4.8.21 Construction and assembly tests

4.8.21.1 Blackout. The shelter assembly under test shall be placed in an area of complete darkness, connected electrically to the appropriate power source and with the lights energized. With all apertures closed and latched, if applicable, and all internal light fixtures operating, no light shall be visible from outside the shelter in any direction. This shall be verified by an observer from all sides including the top. If so equipped, the blackout curtain shall be placed in its track. When the doors is used to enter and exit the shelter, a small amount of light may be visible where the curtain meets the floor per Par 3.3.7. Any other visible light shall be recorded.

4.8.21.2 Floor load. The shelter assembly under test shall be subjected to the test method contained in MIL-STD-907, PAR 5.3.6.1.

4.8.21.3 Airtightness. The shelter assembly under test shall be subjected to the test contained in MIL-STD-907, PAR 5.2.2.2 with all openings closed and latched, if applicable, except the port supplying the make-up air.

4.8.22 Adhesive shear. A minimum of five sample adhesive shear specimens shall be prepared and tested daily in accordance with ASTM E-865-82 procedure. The tests shall be conducted at normal temperature, 75° ± 5°F (23.9° ± 2.8°C). Tests shall be considered satisfactory if the shear stress developed exceeds 2,903 psi ($20{,}015{,}600\ N/m^2$). Results of the tensile pull tests shall be recorded and identified as to the day of test and the lot from which the adhesive was taken. Failure of any specimen shall be cause for rejection of all panels bonded with that adhesive lot.

4.9 Test reports. The results of all testing shall be reported and written in compliance with MIL-STD-831 fully describing and illustrating the test procedures and test results.

5. PACKAGING

5.1 Preservation. The shelter assembly requires no additional preservation.

5.2 Packing. As the shelter assembly is a 20-ft, 1C ISO container, no protective media need be installed prior to shipment. Conventional methods of blocking, bracing and strapping are to be employed during shipment by rail, road and air as described in MIL-STD-1186. Removable markings shall be affixed in accordance with MIL-STD-129 to show cube, weight, and destination specified

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by the contracting agency. When feasible, shelter assemblies shall be shipped in stacks of four coupled together with appropriate vertical connectors.

5.3 Marking. In addition to the marking required by 3.10, each shelter assembly shall be marked for shipment in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. The shelter assembly covered by this specification is one of a family of shelters intended for use by air and ground U.S. Fleet Marine Forces to house and support tactical military equipment and operations in the field.

6.2 Ordering data

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. First article testing required (see 3.1.2).

c. Blackout curtain required (see 3.3.7).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be devlivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are involked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Para No.	Data Requirement Title	Applicable DD No.	Option
4.9.	Test procedures and reports	DI-T-4909	N/A

(Data item descriptions related to this specification and identified in section 6 will be approved and listed as such in DoD 5000.19L, Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

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6.3 First article. The first articles shall be examined and tested for approval at the contractor's plant or at an independent commercial testing laboratory acceptable to the procuring activity. First article tests shall be witnessed by a Government representative of the procuring activity. Approval of the first articles shall be by the procurement contracting officer.

6.4 Quality conformance. When a quality conformance inspection is required, the items to be tested should be a single shelter assembly from each shipping lot. No lot should be accepted until the quality conformance tests have been successfully completed as required by 4.5.

6.5 Subject Term (Keyword) Listing.

Assembly, Shelter, 20 Foot, Rigid
Complexing
Honeycomb
Pannels, Sandwich
Transportable

6.6 International standardization agreements. Certain provisions of this specification (3.1) are the subject of international standardization agreement ISO/DIS 668 and are regulated by the Coast Guard. When amendment, revision or cancellation of the specification is proposed which will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels including departmental standardization offices to change the agreement or make other appropriate accommodations.

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy-MC
Project No. 5411-N038

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-29410B(MC)	Shelter Assembly, 20 Foot Rigid, General Purpose

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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